

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - IV

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Class : SY IT	Batch : 1st
Date of Assignment : 15/07/26	Date/Time of Submission : 15/07/26

Queue Application:

- a. Write a program to: a. Implement a queue using an array.

Code:

```
from array import array

queue = array('i') # Integer array

# Enqueue
queue.append(10)
queue.append(20)
queue.append(30)

print("Queue:", queue)

# Dequeue
print("Deleted:", queue[0])
queue.pop(0)

print("Queue after Dequeue:", queue)

print("Amit 44")
```

```

from array import array

queue = array('i') # Integer array

# Enqueue
queue.append(10)
queue.append(20)
queue.append(30)

print("Queue:", queue)

# Dequeue
print("Deleted:", queue[0])
queue.pop(0)

print("Queue after Dequeue:", queue)

print("Amit 44")

```

Output:

```

Queue: array('i', [10, 20, 30])
Deleted: 10
Queue after Dequeue: array('i', [20, 30])
Amit 44

```

- b. Simulate a simple queuing system (e.g., customer service queue).

Code:

```

from array import array

# Create Queue
customers = array('i')

# Add Customers
customers.append(101)
customers.append(102)
customers.append(103)

```

```
print("Customers in Queue:", customers)
```

```
# Serve First Customer
```

```
print("Served Customer:", customers[0])
```

```
customers.pop(0)
```

```
print("Queue After Serving:", customers)
```

```
print("Amit 44")
```

```
from array import array
```

```
# Create Queue
```

```
customers = array('i')
```

```
# Add Customers
```

```
customers.append(101)
```

```
customers.append(102)
```

```
customers.append(103)
```

```
print("Customers in Queue:", customers)
```

```
# Serve First Customer
```

```
print("Served Customer:", customers[0])
```

```
customers.pop(0)
```

```
print("Queue After Serving:", customers)
```

```
print("Amit 44")
```

Output:

```
Customers in Queue: array('i', [101, 102, 103])
```

```
Served Customer: 101
```

```
Queue After Serving: array('i', [102, 103])
```

```
Amit 44
```

```
|
```

